



OÉ Gaillimh
NUI Galway

Autumn Examinations 2010/ 2011

Exam Code(s) 1MF1, 3BA1, 3BS9, 3EL2, 4BS2, 4CS2,
Exam(s) 3rd & 4th Science, H. Dip. (Appl. Sc), M.Sc.
Software Design & Development

Module Code(s) MP307
Module(s) Modelling II

Paper No. 1
Repeat Paper Yes

External Examiner(s) Professor D.C.G. McKeon
Internal Examiner(s) Professor M. Destrade
Dr P.T. Piironen

Instructions: For full marks answer question 1 in Section A **and**
TWO questions in Section B correctly.
All questions carry the same weight.

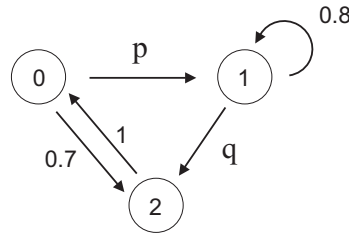
Duration 2hrs
No. of Pages 4
Department(s) Applied Mathematics
Course Co-ordinator(s) Dr P.T. Piironen

Requirements:

MCQ
Handout
Statistical/ Log Tables Yes
Cambridge Tables
Graph Paper
Log Graph Paper
Other Materials

Section A - Answer all questions

1. (a) Describe what is meant by a finite ergodic Markov process with transition probability matrix P . Define the equilibrium probabilities Π_i for a finite ergodic system and show that they obey the matrix equation $\Pi = \Pi P$. Describe a sufficient condition for P which guarantees ergodicity.
- (b) Consider the following Markov Process.



- (i) Find the transition matrix P and the probabilities p and q .
 - (ii) Is the Markov process ergodic?
 - (iii) Find the equilibrium probabilities.
- (c) Consider the nonlinear population model

$$P_{n+1} = R(P_n)P_n$$

where

$$R(P) = \frac{rP}{1 + \left(\frac{P}{K}\right)^2}, \quad r, K > 0.$$

Find the fixed points for this model and show for what initial populations P_0 the population will stabilize and when it will die out .

- (d) Consider the linear control system

$$\begin{aligned} \frac{dx_1}{dt} &= x_2 + u, \\ \frac{dx_2}{dt} &= x_1 + x_2, \\ y &= x_1, \end{aligned}$$

where x_1 and x_2 represent the state, $u \in \mathbb{R}$ is the input and $y \in \mathbb{R}$ is the output.

- (i) Is this system observable?
- (ii) Is this system reachable?

Section B - Answer any **two** questions.

2. (a) Consider a telephone system exchange with r operators. Assuming the probability of a call arriving is independent of the queue size, find the equilibrium probabilities.
- (b) An exchange consists of two operators each of which can deal with 3 customer enquiries per 15 minutes on average. If 180 calls per hour arrive on average what is the equilibrium probability for all lines to be busy? What would be the effect of introducing a third operator?
- (c) On a sunny day, on average, an ice cream salesman services 2 customers every minute with 20 customers arriving every hour. Using a Poisson model for the arrival and servicing mechanism, find the equilibrium probability that more than 3 customers are queuing. What is the average size of the queue? How would the queue change if 30 customers arrived per hour?
3. Consider an ecological system with a prey population of size $x(t)$ and a predator population of size $y(t)$, which satisfy the differential equations

$$\begin{aligned}\frac{dx}{dt} &= (1 - 2y)x, \\ \frac{dy}{dt} &= (-1 + 2x)y.\end{aligned}$$

- (a) Give an interpretation for the various terms in these equations.
- (b) Show that the equilibrium points are $(x, y) = (0, 0)$ and $(x, y) = (1/2, 1/2)$. For small deviations away from the equilibrium point $(x, y) = (1/2, 1/2)$, show that the system is periodic with period π .
- (c) Show that in general the solution satisfies

$$K = \frac{e^{2(x+y)}}{xy}$$

for some constant K . Sketch a plot of x versus y and comment on the typical behaviour of this system in an application of this model.

4. A two-compartment model for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ 0 \end{pmatrix} u, \quad y = c_2,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, p_0 , p_1 , p_2 and b_0 are rate constants, u is the flow rate of the drug into compartment 1 and y is the concentration of the drug in compartment 2.

- (a) Draw a schematic diagram of this system including concentrations, flow directions and rate parameters.
- (b) Investigate for what values of p_0 , p_1 and p_2 this system is observable.
- (c) Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1 , k_2 , k_r are the control parameters, such that the desired output $r = 1$ and the closed loop eigenvalues for the system satisfies $\lambda_1 = -1$ and $\lambda_2 = -2$.